Campus & Hostel Essential Services Finder

Find. Access. Feel at Home.

College Project Submission · Next.js Web Application · 2026

1. Overview

The **Campus & Hostel Essential Services Finder** is a student-focused website designed to make campus life easier. It brings essential campus and hostel services into one place, allowing students to search for facilities, view important details, and find locations conveniently. The goal is to reduce confusion, save time, and help new students feel comfortable in their college environment.

New students often struggle to find the mess, library, medical room, stationery shop, transport, Wi-Fi support, laundry and administration office. This app solves that with a single, searchable directory.

2. Features

- **Homepage** — hero with live search, 10 quick category cards, popular services, important contacts, "How It Works" and an about section.
- **Explore Services** — live search by name/keyword, filter by category and by campus/hostel location, with a friendly "No services found" state and clear filters.
- **Service Details** — each service has its own page with description, location, timings, contact, facilities, availability status and a "Get Directions" link.
- **Campus Map** — an illustrative campus layout with clickable markers and a service list; "Get Directions" opens OpenStreetMap (demo map, clearly labelled).
- **Admin Dashboard** — secure login, add/edit/delete services, activate/deactivate, and live stats (total services, categories, active, inactive).
- **Contact / Help** — validated form in demonstration mode (no real email is sent).
- Fully responsive, clean white + light-blue design with navy headings, rounded cards, soft shadows and lucide-react icons.

3. Tech Stack

Layer	Technology
Frontend	Next.js 15 (App Router), React 19
Backend	Next.js API routes (Node.js)
Database	Supabase (PostgreSQL) when configured; local JSON fallback for instant demo
Auth	bcryptjs password hashing + signed httpOnly session cookie
Icons	lucide-react
Maps	Illustrative SVG layout + OpenStreetMap links
Styling	Plain CSS (no framework)

4. Architecture

The app follows a clean client–server structure within a single Next.js project:

- **Pages** (in `app/`) render the UI — Home, Explore Services, Service Details, Campus Map, About, Contact, Admin Login and Admin Dashboard.
- **API routes** (in `app/api/`) provide the backend: list/search/filter services, create/update/delete services, admin login/logout, stats and contact.
- **Data layer** (`lib/db.js`) reads from Supabase when configured, otherwise from a local JSON file, so the app runs instantly for a demo.
- **Auth layer** (`lib/auth.js`) hashes the admin password with bcrypt and issues a signed httpOnly session cookie; all admin actions are protected on the backend.

5. Setup & Run Instructions

Run locally:

```
npm install
npm run dev
```

Open `http://localhost:3000`.

Default admin login (demo): username `admin`, password `admin123`.

Production database (optional): create a free Supabase project, run `supabase/schema.sql`, then set `SUPABASE_URL` and `SUPABASE_ANON_KEY` in `.env.local`.

Deploy: push to GitHub, import into Vercel (framework Next.js), add the environment variables from `.env.example`, and deploy.

6. Conclusion

This project delivers a complete, working web application — not just a mockup — with functional search, filters, service details, a campus map, database storage and protected admin features. It is intentionally kept simple enough for a first-year engineering student to understand and explain during a college presentation, while still demonstrating real full-stack skills.

> **Assumptions:** sample timings, locations, phone numbers and map links are demonstration data only and are clearly labelled; the campus map is an illustrative demo layout; the contact form runs in demonstration mode and does not send a real email.